Trento, Italy
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Elia Cunegatti

Applied AI Researcher | Efficient LLMs

Website: eliacunegatti.github.io
Github: eliacunegatti

Final-year Ph.D. researcher specializing in efficient large language models, model compression, and post-training optimization. My research focuses on developing novel pruning and quantization methods and building scalable evaluation pipelines for large language models. Industry experience in applied LLM systems at Bristol Myers Squibb, with an incoming Applied Scientist internship at Amazon Alexa AI.

Industry Experience

- **Applied Scientist Intern** September 2026 – March 2027
  Amazon Alexa AI — Turin, Italy

- **Visiting Scientist** April 2026 – June 2026
  Bristol Myers Squibb (CITRE) — Seville, Spain

  - Developed an LLM-based framework to extract gene regulatory interactions from scientific literature and construct gene regulatory networks.
  - Evaluated proprietary and open-source LLMs using prompt engineering, OpenAI APIs, and Amazon Bedrock, building reproducible experimentation pipelines with Docker and SLURM.
  - Collaborated with computational biologists to validate extracted interactions and presented research findings to scientific stakeholders.

Academic Experience

- **Ph.D. Researcher in Efficient Deep Learning** Nov 2022 – Present
  University of Trento, Italy

  - Led and supervised research on model compression via sparsity and quantization, over large-scale AI models (CNNs, LLMs, VLMs, Reinforcement-Learning), resulting in multiple publications.
  - Designed and implemented two state-of-the-art compression frameworks: NeuronAl for unstructured and 2SSP for structured pruning to 60–80% compression on LLMs with minimal accuracy loss.
  - Built scalable experimentation pipelines on HPC clusters (SLURM on NVIDIA A100 GPUs), automating data loading, training (PEFT) and inference using Parallelism, logging, and evaluation for hundreds of runs.
  - Collaborated with external partners such as: Cisco Research, Vrije Universiteit Amsterdam, MHUG, and HiCrest Research Group).

- **Visiting Ph.D. Researcher** Aug 2025 – Oct 2025
  Vrije Universiteit Amsterdam, The Netherlands

  - Investigated how sparsity patterns in Reinforcement-Learning can be used for knowledge-disentangle in controller-based Continual/Meta Reinforcement Learning (still ongoing project).

- **Teaching Assistant & Student Supervisor** Feb 2023 – Present
  University of Trento, Italy

  - Teaching assistant for BSc/MSc courses in Introduction to Machine Learning, Bio-Inspired AI, Optimization Techniques, and Computer Architectures.
  - Supervised 7+ BSc/MSc theses on compression, contrastive Hebbian learning, and multi-objective influence maximization.

- **Research Intern (Exchange Student)** Mar 2022 – Jul 2022
  University of Twente, The Netherlands

  - Developed evolutionary algorithms for large-scale influence maximization on real-world networks, using Python, NetworkX, and multi-objective optimization libraries.
  - Scaled experiments to graphs with millions of edges, profiling and optimizing runtime and memory usage.

Selected Publications

Full publication list on Google Scholar

From Layers to Submodules: Rethinking Granularity in Replacement-Based LLM Compression
Elia Cunegatti, Marcus Vukojevic, Erik Nielsen, Giovanni Iacca
EMNLP 2026 Findings

Frequency Matters: Fast Model-Agnostic Data Curation for Pruning and Quantization
Francesco Pio Monaco, **Elia Cunegatti**, Flavio Vella, Giovanni Iacca
EMNLP 2026 Main

2SSP: A Two-Stage Framework for Structured Pruning of LLMs
Fabrizio Sandri*, **Elia Cunegatti***, Giovanni Iacca * *equal contribution*
*Transactions on Machine Learning Research (**TMLR**), 2025 & **SLLM-Workshop @ ICLR 2025***
Two-stage structured pruning framework (width + depth pruning) for LLMs that jointly removes neurons and attention submodules, automatically balancing sparsity across stages and outperforming five pruning baselines on multiple LLM families and downstream tasks while providing real memory reduction and inference speed-up.

Zeroth-Order Adaptive Neuron Alignment Based Pruning without Re-Training
Elia Cunegatti, Leonardo Lucio Custode, Giovanni Iacca
*Transactions on Machine Learning Research (**TMLR**), 2025 & **SLLM-Workshop @ ICLR 2025***
State-of-the-art top-up unstructured pruning algorithm (NeuronAL) for LLMs that select block-wise and row-wise sparsity using a novel metric called neuron-alignment. Hyperparameter-adaptive and training-free method outperforms, being 20× faster while achieving lower performance-task drop compared to the baselines.

MULTIFLOW: Shifting Towards Task-Agnostic Vision-Language Pruning
Matteo Farina, Massimiliano Mancini, **Elia Cunegatti**, Gaowen Liu, Giovanni Iacca, Elisa Ricci
***CVPR** 2024*
First task-agnostic pruning framework for vision–language models that leverages multimodal information flow and gradient-free saliency to produce a single pruned VLM transferable across multiple downstream tasks.

Understanding Sparse Neural Networks from their Topology via Multipartite Graph Representations
Elia Cunegatti, Matteo Farina, Doina Bucur, Giovanni Iacca
*Transactions on Machine Learning Research (**TMLR**), 2024*
Introduces a multipartite-graph view of sparse networks to analyze connectivity patterns, showing how different pruning strategies induce distinct topologies that correlate with generalization and robustness.

Education

Ph.D. in Computer Science — Nov 2022 – Present
University of Trento, Italy
Thesis focus: Efficient and Sustainable Deep Learning.

M.Sc. in Computer Science, Data Science track — Sep 2020 – Oct 2022
University of Trento, Italy
Final grade: **110/110 cum Laude**.

Erasmus+ Exchange, M.Sc. Computer Science — Aug 2021 – Feb 2022
University of Twente, The Netherlands
GPA: **3.9/4.0**.

B.Sc. in Information Engineering and Business Organization — Sep 2017 – Sep 2020
University of Trento, Italy
Final grade: **107/110**.

Technical Skills

Programming & Tools	Python, Bash, SLURM, Git
Deep Learning & GenAI	PyTorch, HuggingFace, NeuralMagic inference acceleration (CPU/GPU), FSDP/DeepSpeed
Machine Learning & Data	NumPy, Pandas, scikit-learn, Matplotlib, Numba
Optimization & AI	Evolutionary algorithms (DEAP, Inspyred), graph analytics (NetworkX, igraph)
Collaboration & Research	Open-source development, research writing, student supervision, conference presentations
Soft Skills	Teamwork, clear communication, project ownership, mentoring

Honors and Awards

Cisco Scholarship - 100K (awarded with Matteo Farina)	May 2023
Scholarship for attending ICLR 2023 Conference	May 2023
Scholarship for Research Abroad, University of Trento	Mar 2022
Erasmus+ Scholarship, European Union & University of Trento	Mar 2021
Community Interest Degree Award	2019, 2021
University of Trento Merit Scholarship	2019, 2020, 2021, 2022

Languages

Italian	Native
English	C1 (advanced working proficiency)